FEATURE FILM PRODUCTION // MSR CREATIONS // HYDERABAD

THE ART OF INVISIBLE VFX

On-set technical coordination, 3D camera tracking, and photorealistic CG environment integration for the feature film TAKSHAKA.

01 ON-SET TECHNICAL COORDINATION
Live camera calibration, 360° HDR brackets, and marker grid survey.

02 SYNTHEYES 3D CAMERA SOLVE
1,420 point tracking cloud solved with 0.38px reprojection error.

03 CG ENVIRONMENT & LEVEL DESIGN
Unreal Engine level design previs and Maya volumetric set extension.

BENNY JOEL | PRODUCTION COORDINATOR & ASSISTANT VFX SUPERVISOR
DEPARTMENT OF MULTIMEDIA & ANIMATION • FEATURE FILM: TAKSHAKA (MSR CREATIONS)



PLATE TAKSHAKA_CAVE_EXT_V02 — CONFORMED 4K VOLUMETRIC EXTENSION MSR CREATIONS // 2026

On-Set Camera Log & Field Survey

Soundstage calibration notes, lens distortion matrices, and HDR bracket passes on Stage 03.

SECTION 01: OPTICAL CALIBRATION & SENSOR TELEMETRY

Camera Body & Sensor	ARRI ALEXA Mini LF // 4.5K Open Gate (4448×3096)
Active Focal Length	Zeiss Master Prime 35.24mm (T1.3 at f/2.8)
Sensor Line-up / Mount	LPL Mount // Ground Glass Reticle 2.39:1 Anamorphic Ext.
Optical Height & Tilt	Nodal Center: 142.5 cm // Pitch: -8.4° // Roll: +0.2°
Timecode & Shutter	TC 14:22:08:12 // 180.0° Shutter (1/48s) // ISO 800

SECTION 02: PHOTOMETRIC GROUND TRUTH & SURVEY DATA

HDRI Dynamic Range	360° Spherical Panoramic // 7-Bracket RAW (1/8000s to 1/2s)
Chrome & Grey Spheres	Matched to practical stage tungsten & cool rim spill
Color Verification	X-Rite Macbeth ColorChecker v2.4 conformed to ACEScg
Stage Marker Matrix	50mm Fluo Green Crosses on calibrated 100cm floor grid
LiDAR Environment	FARO Focus S350 point cloud registered to millimeter scale

VFX SUPERVISOR FIELD VERIFICATION SIGN-OFF

I hereby verify that all camera parameters, tracking marker placements, and HDR bracket passes for Stage 03 were recorded under director supervision. Boundary cutoffs were negotiated to ensure zero actor overlap with digital cavern extension boundaries.

SUPERVISING ARTIST: BENNY JOEL — STATUS: APPROVED FOR MATCHMOVE // MSR CREATIONS



FIG 02. PRINCIPAL PHOTOGRAPHY // STAGE 03 LIVE BLOCKOUT (TAKE 04)

Live actors positioned within physical rock boundary; background cavern opening demarcated for seamless 3D set extension.

Live practical lighting: Warm 3200K tungsten spotlight on actor, cool ambient backlight to match moonlight.

FEATURE FILM: TAKSHAKA // PRODUCTION PIPELINE DECONSTRUCTION

From Soundstage Sensor to Final Cinema

Progressive breakdown of Shot 047 across raw capture, 3D matchmove, geometry alignment, and volumetric finishing.



01 RAW SENSOR ACQUISITION

ARRI ALEXA LF 4.5K Open Gate raw camera plate. Soundstage ceiling lights and floor tracking markers preserved for spatial registration.

LogC3 // Zeiss 35mm Prime // 24.00 fps



02 3D CAMERA TRACKING

Syntheyes Pro 2026 feature tracking solve. 1,420 multi-frame optical trackers calculated across 240 frames with 0.38px residual error.

Focal: 35.24mm // Residual: 0.38 px RMS



03 SPATIAL GEOMETRY FIT

Metahuman digital double & stasis chamber line-up. Used for physical shadow casting, occlusion, and cryogenic liquid refraction boundaries.

Quad-dominant mesh // Maya (.ma) Conform



04 VOLUMETRIC COMPOSITING

Maya, Unreal Engine 5 & DaVinci Resolve conform. Cavern matte extension, dynamic light shafts, and photometric fog matching tungsten stage spill.

ACEScg // DCI-P3 4K // 100% Director Sign-off

STAGE 03 LIVE CAPTURE

4,448 × 3,096 RAW

ARRI Alexa Mini LF Open Gate calibrated sensor plate.

CAMERA TRACK ACCURACY

0.38 PX RESIDUAL

Sub-pixel reprojection error across 240 active cinematic frames.

CONFORM SIGN-OFF

100% STUDIO APPROVAL

Zero optical drift or edge jitter verified in DaVinci Resolve 4K QC.

FEATURE PRODUCTION: TAKSHAKA // SUPERVISED BY BENNY JOEL — ASST. VFX SUPERVISOR // MSR CREATIONS

Sub-Pixel Matchmove & Asset Fit



SYNTHEYES 3D CAMERA SOLVE — CAMERA01 AT 24.00 FPS | 35.24MM SPHERICAL PRIME | TAKSHAKA SHOT 047

0.38 px

REPROJECTION RESIDUAL ERROR // 1,420 TRACKERS CONVERGED



METAHUMAN MATCHMOVE BUST

Used for exact spatial line-up, shadow casting, and collision boundaries on live actor.

- **Quad-dominant topology**
- **Real-world scale conform**
- **ARRI LF sensor match**

PRODUCTION PIPELINE SPECS // TAKSHAKA CONFORM

Primary Solver	Syntheyes Pro 2026 (Optical Feature Track)
Sensor Gate	ARRI LF OpenGate (4448×3096) @ 24.00 fps
Lens Distortion	3DEqualizer Anamorphic / Radial 4-order
DCC Hand-off	Autodesk Maya (.ma) + The Foundry Nuke (.nk)
Deliverable QC	DaVinci Resolve 10-bit Apple ProRes 4444

CG Environment Integration & Digital Asset Fit

Unreal Engine level design previsualization, Maya 3D environment dressing, and volumetric set extension conformed to camera optics.



FIG 05A. SPATIAL GEOMETRY & CHARACTER ALIGNMENT // SHOT 073

Metahuman digital double and cryo-chamber collision geometry conformed to solved camera path.

Maya (.ma) Rig // Sub-Pixel Collision Mesh // ARRI LF Lens Distortion Calibrated



FIG 05B. VOLUMETRIC SET EXTENSION // FINAL CONFORM PASS

Digital cavern matte extension, dynamic light shafts, and atmospheric haze matching stage depth.

ACEScg Color Space // Exponential Volumetric Fog // 35.24mm Camera Frustum Lock // 100% Sign-Off

I. UNREAL ENGINE LEVEL DESIGN & PREVIS

Authored real-time 3D environment blockouts and spatial level design in Unreal Engine 5 to previsualize cavern dimensions, sightlines, and boundary cutoffs with director.

II. MAYA CG ENVIRONMENT & SHADER INTEGRATION

Supervised 3D environment asset integration in Maya, matching photorealistic rock shaders, displacement, and photometric 3200K tungsten light falloff to live-action plates.

III. VOLUMETRIC ATMOSPHERE & DA VINCI CONFORM

Simulated Rayleigh particulate scattering and dynamic light shafts matching live soundstage haze, delivering 10-bit Apple ProRes 4444 conformed master plates in ACEScg.

Practical Reality vs. Digital Illusion

Direct comparison of on-set stage limitations and the final depth-matched 3D matte extension.



FIG 06A. RAW CAMERA PLATE // OVERHEAD RIGS & MARKERS

Exposed soundstage ceiling, studio lights, and physical ground tracking tape requiring rotoscope & roto-paint.



FIG 06B. FINAL COMPOSITED SHOT // VOLUMETRIC EXTENSION

Cavern geometry, realistic light shafts, and depth-cued particulate haze matching live camera lens.

I. BOUNDARY SURVEY

Negotiated physical stage cutoff lines with the director on set to maximize practical actors while minimizing complex rotoscoping and clean-plate generation.

II. HDRI LIGHT CONFORM

Acquired 360° dynamic range brackets and chrome sphere passes to reproduce warm tungsten stage spill in digital shaders with matching photometric decay.

III. ZERO ARTIFACT SIGN-OFF

Delivered final 4K plate conformed to DaVinci Resolve grading workflow with 100% director sign-off and flawless sub-pixel edge convergence.